HOMEWORK SOLUTION 1

Lec 1

Exercise 1. Compute the empirical and generator-based estimators of one-year transition matrix for the following data.
(a) There are four credit classes, A, B, C, D. At the beginning, there are 20 firms in A, B, C, each.
(b) In one month, one B rated firm migrates to C , and two B rated firms migrate to A , and both firms stay there for the rest of the year.
(c) In three months, one A rated firm migrates to B, and stays there for the rest of the year.
(d) In seven months, two C rated firms migrate to B and stay there for the rest of the year. Furthermore, one C rate firm defaults.
(e) Ten months later, two C rated firms default.
Solution.

$$\hat{P}_{empirical} = \begin{pmatrix} \frac{19}{20} & \frac{1}{20} & 0 & 0 \\ \frac{2}{20} & \frac{17}{20} & \frac{1}{20} & 0 \\ 0 & \frac{2}{20} & \frac{15}{20} & \frac{3}{20} \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

We compute generator-based estimators as follows,
$N_A = 2 \times \frac{12-1}{12} + 1 \times \frac{3}{12} + (20-1) \times \frac{12}{12} = \frac{253}{12}$
$N_B = 3 \times \frac{1}{12} + 1 \times \frac{12-3}{12} + 2 \times \frac{12-7}{12} + (20-3) \times \frac{12}{12} = \frac{226}{12}$
$N_C = 1 \times \frac{12-1}{12} + 3 \times \frac{7}{12} + 2 \times \frac{10}{12} + (20-3-2) \times \frac{12}{12} = \frac{232}{12}$
$M_{A,B} = 1, M_{A,C} = M_{A,D} = 0$
$M_{B,A} = 2, M_{B,C} = 1, M_{A,D} = 0$
$M_{C,A} = 0, M_{C,B} = 2, M_{C,D} = 3$
The estimator becomes

$$\begin{aligned}
\hat{\lambda}_{A,B} &= \frac{M_{A,B}}{N_A} = \frac{12}{253} & \hat{\lambda}_{A,C} &= \hat{\lambda}_{A,D} = 0 \\
\hat{\lambda}_{B,A} &= \frac{M_{B,A}}{N_B} = \frac{24}{226} & \hat{\lambda}_{B,C} &= \frac{M_{B,C}}{N_B} = \frac{12}{226} & \hat{\lambda}_{B,D} &= 0 \\
\hat{\lambda}_{C,B} &= \frac{M_{C,B}}{N_C} = \frac{24}{232} & \hat{\lambda}_{C,D} &= \frac{M_{C,D}}{N_C} = \frac{36}{232} & \hat{\lambda}_{C,A} &= 0
\end{aligned}$$

The estimator of the generator matrix is given by

$$\hat{\Lambda} = \begin{pmatrix} -\frac{12}{253} & \frac{12}{253} & 0 & 0 \\ \frac{24}{226} & -\frac{36}{226} & \frac{12}{226} & 0 \\ 0 & \frac{24}{232} & -\frac{60}{232} & \frac{36}{232} \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Therefore the one-year transition matrix is estimated by

$$\hat{P}_{generator} = e^{\hat{\Lambda}} = \begin{pmatrix} 0.955993091 & 0.0428685943 & 0.00108024669 & 0.0000580677761 \\ 0.0959801271 & 0.857247550 & 0.0431801720 & 0.00359215070 \\ 0.00471211056 & 0.0841268869 & 0.774310356 & 0.136850647 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Exercise 2. A company has issued 3- and 5-year bonds with a coupon payment of 4% per year payable annually. The yields on the bonds (in terms of continuous compounding) are 4.5% and 4.75%, respectively. Assume the risk-free rate is 3.5% for all maturities, and the recovery value is 40% of the par value. Also assume default can only take place halfway through each year, i.e. at year 0.5, 1.5, 2.5, etc. Compute P_1 and P_2, which are the implied default intensities per year for years 1 to 3 and for years 4 and 5, respectively.

Solution For 3 years:

Time (Years)	Default Intensity (per year)	Recovery Value ($)	Risk-Free Value ($)	Loss Given Default ($)	Discount Factor	PV of Expected Loss ($)
0.5	P_1	40	103.0463	63.0463	0.9827	$61.9555P_1$
1.5	P_1	40	102.6456	62.6456	0.9489	$59.4444P_1$
2.5	P_1	40	102.2305	62.2305	0.9162	$57.0156P_1$
					Total:	$178.4155P_1$

Now since the yield is 4.5% the corporate bond price is

$$4e^{-0.045*1} + 4e^{-0.045*2} + 104e^{-0.045*3} = 98.3462$$

Similarly, the default-free bond price is

$$4e^{-0.035*1} + 4e^{-0.035*2} + 104e^{-0.035*1} = 101.2257$$

Setting the difference between the corporate bond and default-free bond equal to the PV of the expected loss yields

$$101.2257 - 98.3462 = 178.4155P_1,$$

i.e. the implied default intensity (or implied one-year default probability given no default yet) is

$$P_1 = 3.6721\%$$

For 5 years:

Time (Years)	Default Intensity (per year)	Recovery Value ($)	Risk-Free Value ($)	Loss Given Default ($)	Discount Factor	PV of Expected Loss ($)
0.5	P_2	40	103.8069	63.8069	0.9827	$62.7030P_2$
2.5	P_2	40	103.0463	63.0463	0.9489	$59.8246P_2$
1.5	P_2	40	103.4332	63.4332	0.9162	$58.1175P_2$
3.5	P_2	40	102.6456	62.6456	0.8847	$55.4226P_2$
4.5	P_2	40	102.2305	62.2305	0.8543	$53.1635P_2$
					Total:	$286.5363P_2$

Now since the yield is 4.75% the corporate bond price is

$$4e^{-0.0475*1} + 4e^{-0.0475*2} + 4e^{-0.0475*3} + 4e^{-0.0475*4} + 104e^{-0.0475*5} = 96.2425$$

Similarly, the default-free bond price is

$$4e^{-0.035*1} + 4e^{-0.035*2} + 4e^{-0.035*3} + 4e^{-0.035*4} + 104e^{-0.035*5} = 101.9743$$

Setting the difference between the corporate bond and default-free bond equal to the PV of the expected loss yields

$$101.9743 - 96.2425 = 286.5363P_1,$$

i.e. the implied default intensity (or implied one-year default probability given no default yet) is

$$P_2 = 2.0004\%$$